

BIOS And POST Troubleshooting Theory

Task 1: 5 Common BIOS or POST Error Messages and Causes

1. "CMOS Battery Failure" / "CMOS Checksum Error"

- **Issue:** The CR2032 coin-cell battery on the motherboard has lost voltage (< 2.5V) or died, causing BIOS configuration settings and system time to reset to defaults upon shutdown.

2. "No bootable device found" / "Insert boot media and press any key"

- **Issue:** The BIOS cannot detect a drive with an operating system bootloader. This points to a disconnected SATA/NVMe cable, a failed SSD/HDD, or an incorrect boot order sequence.

3. "CPU Fan Error" / "CPU Fan Speed Detected Error"

- **Issue:** The CPU cooler fan is disconnected, spinning too slowly, or plugged into the wrong header (e.g., plugged into SYS_FAN or AIO_PUMP instead of the primary CPU_FAN header).

4. "Memory Size / Configuration Has Changed"

- **Issue:** A RAM module has been added, removed, or has failed and is no longer detected by the memory controller, prompting the BIOS to reconfigure its memory map.

5. "Keyboard Error or No Keyboard Present"

- **Issue:** The system cannot detect a keyboard during pre-boot initialization, usually caused by a loose USB cable, a damaged USB port, or legacy USB support being disabled in BIOS.

Task 2: 3 Beeps on AMI BIOS (Meaning & 2 Troubleshooting Steps)

- **What 3 Beeps Mean on an AMI BIOS:**
 - In standard American Megatrends (AMI) BIOS beep code definitions, **3 short beeps** indicate a **Base 64K RAM memory failure** (a fault in the first 64 KB of system memory required to initialize the OS).
- **Two Troubleshooting Steps:**
 1. **Reseat and Clean the RAM:** Turn off power, unplug the power cord, remove all RAM modules, gently clean the golden contact pins with an eraser or isopropyl alcohol, and firmly reseat the modules into the primary memory slots (DIMM A2/B2) until both retention clips click.
 2. **Isolate with One Stick at a Time:** If you have multiple RAM sticks, remove all except one stick in Slot 2 (A2) and test booting. Swap sticks to identify whether a specific RAM stick or a specific motherboard DIMM slot is defective.

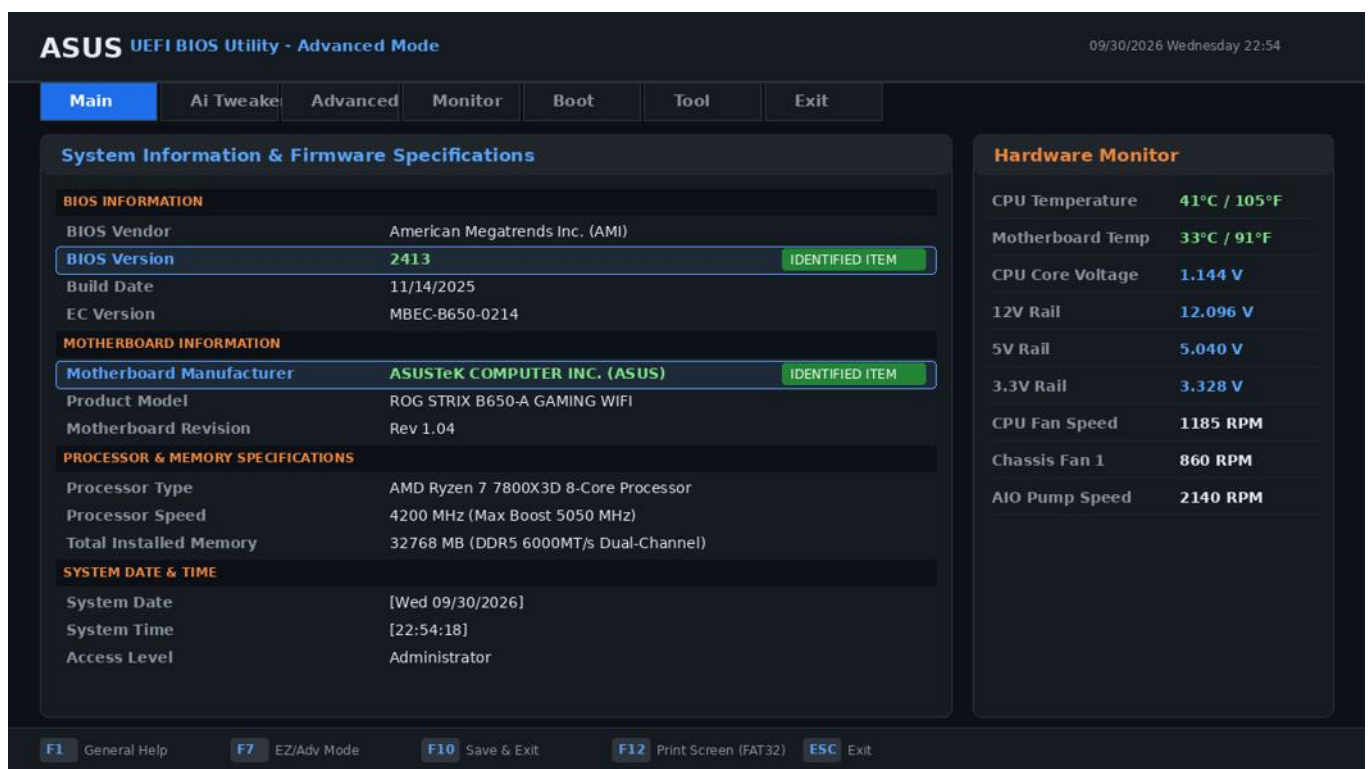
Task 3: Accessing BIOS Setup & Identifying Details

- **How to Access BIOS:**

1. Turn on or restart the PC.
2. Repeatedly tap the setup key (**Del** or **F2** on most desktops; **F10** or **F12** on HP/Dell laptops) immediately when the initial logo appears.
3. Once inside the main screen (EZ Mode or Main Tab), capture a photo or take a screenshot (many modern UEFI BIOS allow saving a screenshot to a FAT32 USB drive by pressing **F12**).

- **Example Details to Identify:**

- **Motherboard Manufacturer / Model:** ASUS / ROG STRIX B650-A GAMING WIFI (or Gigabyte B760 DS3H)
- **BIOS Version:** Version 2413 (Dated 2025/11/14)



Task 4: 'CMOS checksum error – Defaults loaded' Explanation & Fix

- **What It Means:**

- The BIOS checks the integrity of its stored configuration data against a mathematical calculation (a checksum).
- If the values do not match—almost always because the onboard CMOS coin-cell battery lost charge and can no longer supply standby voltage to keep the memory powered—the stored settings become corrupted, and the motherboard automatically loads safe factory default settings.

- **How to Resolve It in a Real System:**

1. **Replace the CMOS Battery:** Disconnect the PC from wall power, locate the round **CR2032 (3V)** coin-cell battery on the motherboard, release the small metal latch to remove the old battery, and insert a fresh CR2032 battery with the positive (+) side facing up.
2. **Reconfigure and Save BIOS Settings:** Boot into the BIOS setup utility, update the correct **Date and Time**, re-enable your boot drive order and memory profile (such as **XMP / EXPO**), and press **F10 (Save and Exit)**. Subsequent reboots will now save settings cleanly without displaying the checksum error.

Phoenix - AwardBIOS v6.00PG, An Energy Star Ally
Copyright (C) 1984-2026, Phoenix Technologies, LTD

EPA POLLUTION
PREVENTER
energy

ASUS PRIME B650M-A ACPI BIOS Revision 1807

Main Processor : AMD Ryzen 5 7600X 6-Core Processor (4.70GHz)
Memory Testing : 16777216K OK

Primary Master : Samsung SSD 980 1TB (NVMe PCIe 4.0)
Primary Slave : None
Secondary Master : None
Secondary Slave : None

USB Devices total: 1 Drive, 1 Keyboard, 1 Mouse, 0 Hub
Auto-detecting USB Mass Storage Devices .. 00 USB mass storage devices found and configured.

CMOS battery failed

CMOS checksum error - Defaults loaded

Warning: System date/time and hardware configuration have been reset.

Press F1 to continue, DEL to enter SETUP
Press F2 to load default values and continue

09/30/2026-B650-ALC897-6A61RM4DC-00